

On the end-point of Stein-Weiss inequality

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Abstract

We prove that the classical Stein-Weiss inequality is true for $p = 1$.

1 Introduction

Let $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. We define

$$\mathbf{I}_{\alpha\gamma\delta}f(x) = \int_{\mathbb{R}^{\mathbf{N}}} f(y) \left(\frac{1}{|x|}\right)^{\gamma} \left(\frac{1}{|x-y|}\right)^{\mathbf{N}-\alpha} \left(\frac{1}{|y|}\right)^{\delta} dy, \quad x \neq 0. \quad (1.1)$$

In 1928, Hardy and Littlewood [1] first investigated the $\mathbf{L}^p - \mathbf{L}^q$ -regularity of $\mathbf{I}_{\alpha\gamma\delta}$ at $\mathbf{N} = 1$. Thirty years later, Stein and Weiss [3] extended this result to every higher dimensional space. Today, it bears the name of Stein-Weiss inequality.

◊ Throughout, $\mathfrak{C} > 0$ is regarded as a generic constant depending on its subindices.

Theorem A: Stein and Weiss, 1958

Let $\mathbf{I}_{\alpha\gamma\delta}$ defined in (1.1) for $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. We have

$$\|\mathbf{I}_{\alpha\gamma\delta}f\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \mathfrak{C}_{\alpha\gamma\delta p q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \quad 1 < p \leq q < \infty \quad (1.2)$$

if and only if

$$\gamma < \frac{\mathbf{N}}{q}, \quad \delta < \mathbf{N} \left(\frac{p-1}{p} \right), \quad \gamma + \delta \geq 0, \quad \frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma + \delta}{\mathbf{N}}. \quad (1.3)$$

Remark 1.1. In the original paper of Stein and Weiss [3], (1.3) is given as a sufficient condition. These constraints of $\alpha, \gamma, \delta, p, q$ in (1.3) are in fact necessary. See subsection 3.1.

When $\gamma = \delta = 0$, **Theorem A** was proved in $\mathbb{R}^{\mathbf{N}}$ by Sobolev [2]. This is also known as Hardy-Littlewood-Sobolev inequality.

The theory of fractional integration in weighted norms has been substantially developed during the second half of 20th century. See Fefferman and Muckenhoupt [6], Muckenhoupt and Wheeden [7], Pérez [10] and Sawyer and Wheeden [8].

Our main result is an improvement of **Theorem A** to include the end-point $p = 1$.

Theorem One Let $\mathbf{I}_{\alpha\gamma\delta}$ defined in (1. 1) for $0 < \alpha < \mathbf{N}$ and $\gamma, \delta < \mathbf{N}$. We have

$$\|\mathbf{I}_{\alpha\gamma\delta}f\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \leq \mathfrak{C}_{\alpha\gamma\delta p q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \quad 1 \leq p \leq q < \infty \quad (1. 4)$$

if and only if

$$\gamma < \frac{\mathbf{N}}{q}, \quad \delta < \mathbf{N}\left(\frac{p-1}{p}\right), \quad \gamma + \delta \geq 0, \quad \frac{\alpha}{\mathbf{N}} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma + \delta}{\mathbf{N}}. \quad (1. 5)$$

Remark 1.2. For $p = 1$ and $\gamma + \delta > 0$, **Theorem One** is recently proved by Nápoli and Picon [9] whereas (1. 4) is equivalently considered as a two-weight $\mathbf{L}^p \rightarrow \mathbf{L}^q$ -norm inequality for $f * |x|^{\alpha-\mathbf{N}}$ by taking into account $|x|^{-\gamma}, |x|^{\delta}$ as the weights. The regarding estimates are developed in analogue to the work of Sawyer and Wheeden [8]. See Section 2 of [9]. In particular, the 'good kernel' approximation used there is nice, but cannot handle the case $\gamma + \delta = 0$.

This paper is organized as follows. The next section is devoted to two fundamental lemmas initially given by Stein and Weiss [3] for $p > 1$. These results are improved now to become applicable for $p \geq 1$. We prove **Theorem One** in Section 3.

2 Two fundamental lemmas

Lemma 2.1. Let $\Omega(u, v) \geq 0$ defined in the quadrant $\{(u, v) : u \geq 0, v \geq 0\}$ which is homogeneous of degree $-\mathbf{N}$ and

$$\mathbf{A} \doteq \int_0^\infty \Omega(1, t) t^{\mathbf{N}(\frac{p-1}{p})-1} dt < \infty. \quad (2. 1)$$

Consider

$$\mathbf{U}f(x) = \int_{\mathbb{R}^{\mathbf{N}}} \Omega(|x|, |y|) f(y) dy. \quad (2. 2)$$

We have

$$\|\mathbf{U}f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \leq \mathfrak{C}_{\mathbf{A}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \quad 1 \leq p < \infty. \quad (2. 3)$$

Proof. Let $R = |x|$ and $r = |y|$. Write $x = R\xi$ and $y = r\eta$ of which ξ, η are unit vectors. We have

$$\mathbf{U}f(x) = \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta) \quad (2. 4)$$

where $d\sigma$ is the surface measure on $\mathbb{S}^{\mathbf{N}-1}$.

In particular for $\mathbf{N} = 1$, $d\sigma$ is the point measure on $\{-1, 1\}$.

Recall that Ω is homogeneous of degree $-\mathbf{N}$. Consider

$$\begin{aligned}
& \left\{ \int_0^\infty \left| \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr \right|^p R^{\mathbf{N}-1} dR \right\}^{\frac{1}{p}} \\
&= \left\{ \int_0^\infty \left| \int_0^\infty \Omega(1, t) f(tR\eta) t^{\mathbf{N}-1} dt \right|^p R^{\mathbf{N}-1} dR \right\}^{\frac{1}{p}} \\
&\leq \int_0^\infty \Omega(1, t) t^{\mathbf{N}-1} \left\{ \int_0^\infty |f(tR\eta)|^p R^{\mathbf{N}-1} dR \right\}^{\frac{1}{p}} dt \quad \text{by Minkowski integral inequality} \quad (2.5) \\
&= \int_0^\infty \Omega(1, t) t^{\mathbf{N}[1-\frac{1}{p}]-1} \left\{ \int_0^\infty |f(r\eta)|^p r^{\mathbf{N}-1} dr \right\}^{\frac{1}{p}} dt \\
&= \mathbf{A} \left\{ \int_0^\infty |f(r\eta)|^p r^{\mathbf{N}-1} dr \right\}^{\frac{1}{p}}.
\end{aligned}$$

For $\mathbf{N} \geq 2$, we find

$$\begin{aligned}
\| \mathbf{U}f \|_{L^p(\mathbb{R}^{\mathbf{N}})} &= \left\{ \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \left| \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta) \right|^p R^{\mathbf{N}-1} dR d\sigma(\xi) \right\}^{\frac{1}{p}} \\
&= \omega_{\mathbf{N}-1}^{\frac{1}{p}} \left\{ \int_0^\infty \left| \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta) \right|^p R^{\mathbf{N}-1} dR \right\}^{\frac{1}{p}} \\
&\leq \omega_{\mathbf{N}-1}^{\frac{1}{p}} \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{ \int_0^\infty \left| \int_0^\infty \Omega(R, r) f(r\eta) r^{\mathbf{N}-1} dr \right|^p R^{\mathbf{N}-1} dR \right\}^{\frac{1}{p}} d\sigma(\eta) \\
&\quad \text{by Minkowski integral inequality} \quad (2.6) \\
&\leq \omega_{\mathbf{N}-1}^{\frac{1}{p}} \mathbf{A} \int_{\mathbb{S}^{\mathbf{N}-1}} \left\{ \int_0^\infty |f(r\eta)|^p r^{\mathbf{N}-1} dr \right\}^{\frac{1}{p}} d\sigma(\eta) \quad \text{by (2.5)} \\
&\leq \mathbf{A} \omega_{\mathbf{N}-1}^{\frac{1}{p}} \left\{ \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty |f(r\eta)|^p r^{\mathbf{N}-1} dr d\sigma(\eta) \right\}^{\frac{1}{p}} \left\{ \int_{\mathbb{S}^{\mathbf{N}-1}} d\sigma(\eta) \right\}^{\frac{p-1}{p}} \\
&\quad \text{by Hölder inequality} \\
&= \mathbf{A} \omega_{\mathbf{N}-1} \|f\|_{L^p(\mathbb{R}^{\mathbf{N}})}
\end{aligned}$$

where

$$\omega_{\mathbf{N}-1} = 2\pi^{\frac{\mathbf{N}}{2}} \Gamma^{-1}\left(\frac{\mathbf{N}}{2}\right)$$

is the area of $\mathbb{S}^{\mathbf{N}-1}$.

When $\mathbf{N} = 1$, $d\sigma$ is the point measure on 1 and -1 . The same estimates hold in (2.6). $\square$

Lemma 2.2. Define $\Delta(t, \xi, \eta) = |1 - 2t\xi \cdot \eta + t^2|^{\frac{1}{2}}$ for $t > 0$ and $\xi, \eta \in \mathbb{S}^{N-1}$. We have

$$\begin{aligned} \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\xi) &= \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\eta) \\ &\leq \mathfrak{C} |1 - t|^{-\frac{N-\alpha}{2}}, \quad t \neq 1, \quad \xi, \eta \in \mathbb{S}^{N-1}. \end{aligned} \quad (2.7)$$

Proof. When $N = 1$, $d\sigma$ is the point measure on 1 or -1 . We find $\Delta(t, \xi, \eta) = |1 \pm t|$, $t > 0$ for $\xi, \eta \in \{-1, 1\}$. The two integrals in (2.7) are clearly bounded by $|1 - t|^{\alpha-1}$ for $t > 0, t \neq 1$.

Let $N \geq 2$. Note that $\Delta(t, \xi, \eta)$ is symmetric *w.r.t* ξ and η . For $0 < t < 1$, we have

$$\mathbf{P}(\xi, t\eta) = \frac{1 - |t\eta|^2}{|\xi - t\eta|^N} = \frac{1 - t^2}{\Delta^N(t, \xi, \eta)}, \quad \xi, \eta \in \mathbb{S}^{N-1} \quad (2.8)$$

which is the Poisson kernel on the unit sphere $\mathbb{S}^{N-1}$.

A direct computation shows

$$\Delta_\eta \mathbf{P}(\xi, t\eta) = 0, \quad \xi \in \mathbb{S}^{N-1} \quad (2.9)$$

where Δ_η is the Laplacian operator *w.r.t* η .

By using the mean value property of harmonic functions, we find

$$1 = \mathbf{P}(\xi, 0) = \frac{1}{\omega_{N-1}} \int_{\mathbb{S}^{N-1}} \mathbf{P}(\xi, t\eta) d\sigma(\eta), \quad 0 < t < 1. \quad (2.10)$$

This further implies

$$\frac{1}{\omega_{N-1}} \int_{\mathbb{S}^{N-1}} \frac{1 - t^2}{\Delta^N(t, \xi, \eta)} d\sigma(\eta) = 1, \quad 0 < t < 1. \quad (2.11)$$

On the other hand, write $0 < s = t^{-1} < 1$ for $t > 1$. From (2.8), we have

$$\begin{aligned} \frac{1 - t^2}{\Delta^N(t, \xi, \eta)} &= \frac{1 - t^2}{|1 + 2t\xi \cdot \eta + t^2|^{\frac{N}{2}}} = \frac{t^2(s^2 - 1)}{t^N |s^2 + 2s\xi \cdot \eta + 1|^{\frac{N}{2}}} \\ &= -t^{2-N} \frac{1 - s^2}{\Delta^N(s, \xi, \eta)} = -t^{2-N} \mathbf{P}(\xi, s\eta), \quad \xi, \eta \in \mathbb{S}^{N-1}. \end{aligned} \quad (2.12)$$

By using (2.10) and (2.12), we find

$$\begin{aligned} \frac{1}{\omega_{N-1}} \int_{\mathbb{S}^{N-1}} \frac{1 - t^2}{\Delta^N(t, \xi, \eta)} d\sigma(\eta) &= -t^{2-N} \frac{1}{\omega_{N-1}} \int_{\mathbb{S}^{N-1}} \mathbf{P}(\xi, s\eta) d\sigma(\eta) \\ &= -t^{2-N}, \quad t > 1. \end{aligned} \quad (2.13)$$

By putting together (2.11) and (2.13), we obtain

$$\frac{1}{\omega_{N-1}} \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^N(t, \xi, \eta)} d\sigma(\eta) \leq \frac{1}{|1 - t^2|} < \frac{1}{|1 - t|}. \quad (2.14)$$

Lastly, by applying Hölder's inequality, we have

$$\begin{aligned}
\int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\eta) &\leq \left\{ \int_{\mathbb{S}^{N-1}} \left[\frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} \right]^{\frac{N}{N-\alpha}} d\sigma(\eta) \right\}^{\frac{N-\alpha}{N}} \left\{ \int_{\mathbb{S}^{N-1}} d\sigma(\eta) \right\}^{\frac{\alpha}{N}} \\
&= \left\{ \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^N(t, \xi, \eta)} d\sigma(\eta) \right\}^{\frac{N-\alpha}{N}} (\omega_{N-1})^{\frac{\alpha}{N}} \\
&\leq \left(\frac{1}{|1-t|} \right)^{\frac{N-\alpha}{N}} (\omega_{N-1})^{\frac{N-\alpha}{N}} (\omega_{N-1})^{\frac{\alpha}{N}} \quad \text{by (2. 14)} \\
&= \omega_{N-1} |1-t|^{-\frac{N-\alpha}{N}}.
\end{aligned} \tag{2. 15}$$

3 Proof of Theorem One

3.1 The $L^p \longrightarrow L^q$ -norm inequality in (1. 4) implies the constraints in (1. 5)

Case 1 Let $p = 1$. We first carry out an implication given by Sawyer and Wheeden [8].

Denote Q as any cube in $\mathbb{R}^N$. Choose $f = \chi_Q$ which is an indicator function supported in Q . The $L^p \longrightarrow L^q$ -norm inequality in (1. 4) implies

$$\sup_{Q \subset \mathbb{R}^N} |Q|^{\frac{\alpha}{N}-1+\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\gamma q} dx \right\}^{\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\delta} dx \right\} < \infty. \tag{3. 1}$$

A standard exercise of changing dilations inside (3. 1) shows that $\frac{\alpha}{N} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{N}$ is an necessary condition. Moreover, it is essential to have $\gamma < \frac{N}{q}$ for the local integrability of $|x|^{-\gamma q}$. We claim $\frac{\alpha}{N} - 1 + \frac{1}{q} \geq 0$. Together with $\frac{\alpha}{N} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{N}$, we must have $\gamma + \delta \geq 0$.

Suppose $\frac{\alpha}{N} - 1 + \frac{1}{q} < 0$. Let Q centered on some $x_o \neq 0$. By shrinking Q to x_o and applying Lebesgue's Differentiation Theorem, we find

$$\left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\gamma q} dx \right\}^{\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|} \right)^{\delta} dx \right\} = |x_o|^{-(\gamma+\delta)} > 0.$$

On the other hand, $|Q|^{\frac{\alpha}{N}-1+\frac{1}{q}} \longrightarrow \infty$. This contradicts to (3. 1).

Recall $\mathbf{I}_{\alpha\gamma\delta}f$ defined in (1. 1). Let $f \geq 0$, $f \in L^1(\mathbb{R}^N)$. Moreover, f is supported in the unit ball, denoted by $\mathbf{B}$.

We have

$$\begin{aligned}
\mathbf{I}_{\alpha\gamma\delta}f(x) &= \int_{\mathbb{R}^N} f(y) \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta dy \\
&\geq \chi(|x| > 10) \int_{\mathbf{B}} f(y) \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta dy \\
&> 2^{\alpha-N} \left(\frac{1}{|x|}\right)^{N-\alpha+\gamma} \chi(|x| > 10) \int_{\mathbf{B}} f(y) \left(\frac{1}{|y|}\right)^\delta dy.
\end{aligned} \tag{3.2}$$

Observe that if $\delta \geq 0$, then $\frac{\alpha}{N} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{N}$ implies $N - \alpha + \gamma \leq \frac{N}{q}$. Consequently, $[\mathbf{I}_{\alpha\gamma\delta}f]^q$ cannot be integrable in $\mathbb{R}^N$. Therefore, we also need $\delta < 0$.

Case 2 Let $p > 1$. Consider $f(x) = \chi_Q(x)|x|^{-\delta(\frac{p}{p-1})}$. As a well known, (1. 4) implies

$$\sup_{Q \subset \mathbb{R}^N} |Q|^{\frac{\alpha}{N} - \frac{1}{p} + \frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|}\right)^{\gamma q} dx \right\}^{\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|}\right)^{\delta(\frac{p}{p-1})} dx \right\}^{\frac{p-1}{p}} < \infty. \tag{3.3}$$

We essentially need $\gamma < N/q$ and $\delta < N(\frac{p-1}{p})$ for which $|x|^{-\gamma q}$ and $|x|^{-\delta(\frac{p}{p-1})}$ are locally integrable. By changing dilations inside (3. 3), we find $\frac{\alpha}{N} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{N}$ is a necessity. Moreover, we claim $\frac{\alpha}{N} \geq \frac{1}{p} - \frac{1}{q}$. Together with $\frac{\alpha}{N} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{N}$, we must have $\gamma + \delta \geq 0$.

Suppose $\frac{\alpha}{N} < \frac{1}{p} - \frac{1}{q}$. Let Q centered on some $x_o \neq 0$. By shrinking Q to x_o and applying Lebesgue's Differentiation Theorem, we find

$$\left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|}\right)^{\gamma q} dx \right\}^{\frac{1}{q}} \left\{ \frac{1}{|Q|} \int_Q \left(\frac{1}{|x|}\right)^{\delta(\frac{p}{p-1})} dx \right\}^{\frac{p-1}{p}} = |x_o|^{-(\gamma+\delta)} > 0.$$

On the other hand, $|Q|^{\frac{\alpha}{N} - \frac{1}{p} + \frac{1}{q}} \rightarrow \infty$. We reach a contradiction to (3. 3).

3.2 The constraints in (1. 5) imply the $L^p \rightarrow L^q$ -norm inequality in (1. 4)

Consider

$$\mathbf{I}_{\alpha\gamma\delta}f(x) = \mathbf{U}_1f(x) + \mathbf{U}_2f(x) + \mathbf{U}_3f(x), \quad f \geq 0 \tag{3.4}$$

where

$$\begin{aligned}
\mathbf{U}_1f(x) &= \int_{\mathbb{R}^N} f(y) \Omega_1(x, y) dy, \\
\Omega_1(x, y) &= \begin{cases} \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta, & 0 \leq \frac{|y|}{|x|} \leq \frac{1}{2}, \\ 0, & \frac{|y|}{|x|} > \frac{1}{2}; \end{cases}
\end{aligned} \tag{3.5}$$

$$\begin{aligned} \mathbf{U}_2 f(x) &= \int_{\mathbb{R}^N} f(y) \Omega_2(x, y) dy, \\ \Omega_2(x, y) &= \begin{cases} \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta, & \frac{|y|}{|x|} \geq 2, \\ 0, & 0 \leq \frac{|y|}{|x|} < 2 \end{cases} \end{aligned} \quad (3.6)$$

and

$$\begin{aligned} \mathbf{U}_3 f(x) &= \int_{\mathbb{R}^N} f(y) \Omega_3(x, y) dy, \\ \Omega_3(x, y) &= \begin{cases} \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta, & \frac{1}{2} < \frac{|y|}{|x|} < 2, \\ 0, & 0 \leq \frac{|y|}{|x|} \leq \frac{1}{2} \text{ or } \frac{|y|}{|x|} \geq 2. \end{cases} \end{aligned} \quad (3.7)$$

We aim to show that each $\mathbf{U}_i f, i = 1, 2, 3$ satisfies the $\mathbf{L}^p \rightarrow \mathbf{L}^q$ -norm inequality in (1.4). The first two can be obtained by refining the proof of Stein and Weiss [3] and using the two lemmas from the previous section. The crucial part of our estimates comes when dealing with $\mathbf{U}_3 f$ for which we go through an interpolation argument of changing measures.

Case One Let $1 \leq p = q < \infty$. The homogeneity condition $\frac{\alpha}{N} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{N}$ implies $\alpha = \gamma + \delta$.

Note that $|x - y| \geq \frac{1}{2}|x|$ if $\frac{|y|}{|x|} \leq \frac{1}{2}$. From (3.5), we find

$$\Omega_1(x, y) \leq 2^{N-\alpha} \begin{cases} \left(\frac{1}{|x|}\right)^{N-\alpha+\gamma} \left(\frac{1}{|y|}\right)^\delta, & 0 \leq \frac{|y|}{|x|} \leq \frac{1}{2}, \\ 0, & \frac{|y|}{|x|} > \frac{1}{2}. \end{cases} \quad (3.8)$$

Because $\delta < N\left(\frac{p-1}{p}\right)$, we have

$$\begin{aligned} \mathbf{A}_1 &\doteq \int_0^\infty \Omega_1(1, t) t^{N\left(\frac{p-1}{p}\right)-1} dt \\ &\leq 2^{N-\alpha} \int_0^{\frac{1}{2}} t^{N\left(\frac{p-1}{p}\right)-\delta-1} dt < \infty. \end{aligned} \quad (3.9)$$

By applying Lemma 2.1, we obtain

$$\|\mathbf{U}_1 f\|_{\mathbf{L}^p(\mathbb{R}^N)} \leq \mathfrak{C}_{\alpha, \delta, p} \|f\|_{\mathbf{L}^p(\mathbb{R}^N)}, \quad 1 \leq p < \infty. \quad (3.10)$$

On the other hand, $|x - y| \geq \frac{1}{2}|y|$ if $\frac{|y|}{|x|} \geq 2$. From (3.6), we find

$$\Omega_2(x, y) \leq 2^{N-\alpha} \begin{cases} \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|y|}\right)^{N-\alpha+\delta}, & \frac{|y|}{|x|} \geq 2, \\ 0, & 0 \leq \frac{|y|}{|x|} < 2. \end{cases} \quad (3.11)$$

Because $\gamma < \mathbf{N}/q = \mathbf{N}/p$ and $\alpha = \gamma + \delta$, we have

$$\begin{aligned} \mathbf{A}_2 &\doteq \int_0^\infty \Omega_2(1, t) t^{\mathbf{N}(\frac{p-1}{p})-1} dt \\ &\leq 2^{\mathbf{N}-\alpha} \int_2^\infty t^{\mathbf{N}(\frac{p-1}{p})-\mathbf{N}+\alpha-\delta-1} dt = 2^{\mathbf{N}-\alpha} \int_2^\infty t^{-\frac{\mathbf{N}}{p}+\gamma-1} dt < \infty. \end{aligned} \quad (3.12)$$

By applying Lemma 2.1, we obtain

$$\|\mathbf{U}_2 f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})} \leq \mathfrak{C}_{\alpha \gamma p} \|f\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \quad 1 \leq p < \infty. \quad (3.13)$$

Write $x = R\xi$ and $y = r\eta$ for $\xi, \eta \in \mathbb{S}^{\mathbf{N}-1}$. Recall Ω_3 defined in (3.7) which is homogeneous of degree $-\mathbf{N}$.

We have

$$\begin{aligned} \mathbf{U}_3 f(x) &= \int_{\mathbb{R}^{\mathbf{N}}} \Omega_3(x, y) f(y) dy \\ &= \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega_3(R\xi, r\eta) f(r\eta) r^{\mathbf{N}-1} dr d\sigma(\eta) \\ &= \int_{\mathbb{S}^{\mathbf{N}-1}} \int_0^\infty \Omega_3(\xi, t\eta) f(tR\eta) t^{\mathbf{N}-1} dt d\sigma(\eta) \\ &= \int_{\mathbb{S}^{\mathbf{N}-1}} \int_{\frac{1}{2}}^2 \frac{1}{|\xi - t\eta|^{\mathbf{N}-\alpha}} f(tR\eta) t^{\mathbf{N}-1-\delta} dt d\sigma(\eta) \quad \text{by (3.7)} \\ &= \int_{\mathbb{S}^{\mathbf{N}-1}} \int_{\frac{1}{2}}^2 \frac{1}{[(\xi - t\eta) \cdot (\xi - t\eta)]^{\frac{\mathbf{N}-\alpha}{2}}} f(tR\eta) t^{\mathbf{N}-1-\delta} dt d\sigma(\eta) \\ &= \int_{\mathbb{S}^{\mathbf{N}-1}} \int_{\frac{1}{2}}^2 \frac{1}{|1 - 2t\xi \cdot \eta + t^2|^{\frac{\mathbf{N}-\alpha}{2}}} f(tR\eta) t^{\mathbf{N}-1-\delta} dt d\sigma(\eta) \\ &\lesssim \int_{\mathbb{S}^{\mathbf{N}-1}} \int_{\frac{1}{2}}^2 \frac{1}{\Delta^{\mathbf{N}-\alpha}(t, \xi, \eta)} f(tR\eta) dt d\sigma(\eta). \end{aligned} \quad (3.14)$$

In particular, for $\mathbf{N} = 1$, $d\sigma$ is a point measure on 1 and -1 inside (3.14). We find

$$\mathbf{U}_3 f(x) \lesssim \int_{\frac{1}{2}}^2 |1 - t|^{\alpha-1} [f(tR) + f(-tR)] dt. \quad (3.15)$$

From (3. 14), we have

$$\begin{aligned}
\|U_3 f\|_{L^p(\mathbb{R}^N)} &\lesssim \left\{ \int_{\mathbb{S}^{N-1}} \int_0^\infty \left\{ \int_{\mathbb{S}^{N-1}} \int_{\frac{1}{2}}^2 \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} f(tR\eta) dt d\sigma(\eta) \right\}^p R^{N-1} dR d\sigma(\xi) \right\}^{\frac{1}{p}} \\
&\lesssim \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty \int_{\mathbb{S}^{N-1}} \left\{ \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} f(tR\eta) d\sigma(\eta) \right\}^p d\sigma(\xi) R^{N-1} dR \right\}^{\frac{1}{p}} dt \\
&\quad \text{by Minkowski integral inequality} \\
&\lesssim \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty \int_{\mathbb{S}^{N-1}} \left\{ \int_{\mathbb{S}^{N-1}} \frac{[f(tR\eta)]^p}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\eta) \right\} \left\{ \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\eta) \right\}^{p-1} d\sigma(\xi) R^{N-1} dR \right\}^{\frac{1}{p}} dt \\
&\quad \text{by Hölder inequality} \\
&\lesssim \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty \int_{\mathbb{S}^{N-1}} \left\{ \int_{\mathbb{S}^{N-1}} \frac{[f(tR\eta)]^p}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\eta) \right\} |1 - t|^{\lfloor \frac{\alpha-N}{N} \rfloor (p-1)} d\sigma(\xi) R^{N-1} dR \right\}^{\frac{1}{p}} dt \\
&\quad \text{by Lemma 2.2} \\
&= \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty \int_{\mathbb{S}^{N-1}} \left\{ |1 - t|^{\lfloor \frac{\alpha-N}{N} \rfloor (p-1)} \int_{\mathbb{S}^{N-1}} \frac{1}{\Delta^{N-\alpha}(t, \xi, \eta)} d\sigma(\xi) \right\} [f(tR\eta)]^p R^{N-1} dR d\sigma(\eta) \right\}^{\frac{1}{p}} dt \\
&\lesssim \int_{\frac{1}{2}}^2 \left\{ |1 - t|^{\lfloor \frac{\alpha-N}{N} \rfloor p} \int_0^\infty \int_{\mathbb{S}^{N-1}} [f(tR\eta)]^p R^{N-1} dR d\sigma(\eta) \right\}^{\frac{1}{p}} dt \quad \text{by Lemma 2.2} \\
&= \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty \int_{\mathbb{S}^{N-1}} [f(tR\eta)]^p d\sigma(\eta) R^{N-1} dR \right\}^{\frac{1}{p}} |1 - t|^{\frac{\alpha-N}{N}} dt \\
&= \|f\|_{L^p(\mathbb{R}^N)} \int_{\frac{1}{2}}^2 |1 - t|^{\frac{\alpha-N}{N}} dt \leq \mathfrak{C}_\alpha \|f\|_{L^p(\mathbb{R}^N)}, \quad 1 \leq p < \infty.
\end{aligned} \tag{3. 16}$$

Moreover, by using (3. 15), we find

$$\begin{aligned}
\|U_3 f\|_{L^p(\mathbb{R})} &\lesssim \left\{ \int_0^\infty \left\{ \int_{\frac{1}{2}}^2 |1 - t|^{\alpha-1} [f(tR) + f(-tR)] dt \right\}^p dR \right\}^{\frac{1}{p}} \\
&\leq \int_{\frac{1}{2}}^2 \left\{ \int_0^\infty [f(tR) + f(-tR)]^p dR \right\}^p |1 - t|^{\alpha-1} dt \\
&\quad \text{by Minkowski integral inequality} \\
&= \|f\|_{L^p(\mathbb{R})} \int_{\frac{1}{2}}^2 |1 - t|^{\alpha-1} dt \\
&\leq \mathfrak{C}_\alpha \|f\|_{L^p(\mathbb{R})}, \quad 1 \leq p < \infty.
\end{aligned} \tag{3. 17}$$

Case Two Consider $1 \leq p < q < \infty$. Assert

$$\mathbf{V}_\delta f(x) = |x|^{-\mathbf{N}+\delta} \int_{|y|<|x|} |y|^{-\delta} f(y) dy, \quad \delta < \mathbf{N} \left(\frac{p-1}{p} \right). \quad (3.18)$$

We claim

$$\|\mathbf{V}_\delta f\|_{\mathbf{L}^p(\mathbb{R}^N)} \leq \mathfrak{C}_{\delta,p} \|f\|_{\mathbf{L}^p(\mathbb{R}^N)}, \quad 1 \leq p < \infty. \quad (3.19)$$

Write

$$\mathbf{V}_\delta f(x) = \int_{\mathbb{R}^N} \Omega(|x|, |y|) f(y) dy, \quad \Omega(u, v) = \begin{cases} u^{-\mathbf{N}+\delta} v^{-\delta} & \text{if } v < u \\ 0 & \text{otherwise.} \end{cases} \quad (3.20)$$

Observe that Ω in (3.20) is homogeneous of degree $-\mathbf{N}$. Moreover,

$$\int_0^\infty \Omega(1, t) t^{\mathbf{N}(\frac{p-1}{p})-1} dt = \int_0^1 t^{\mathbf{N}(\frac{p-1}{p})-\delta-1} dt < \infty \quad (3.21)$$

provided by $\delta < \mathbf{N} \left(\frac{p-1}{p} \right)$. Lemma 2.1 implies (3.19).

On the other hand, $\mathbf{V}_\delta f$ defined in (3.18) satisfies

$$\begin{aligned} \mathbf{V}_\delta f(x) &\leq |x|^{-\mathbf{N}+\delta} \left\{ \int_{|y|<|x|} |y|^{-\delta(\frac{p}{p-1})} dy \right\}^{\frac{p-1}{p}} \|f\|_{\mathbf{L}^p(\mathbb{R}^N)} \quad \text{by Hölder inequality} \\ &\leq \mathfrak{C}_{\delta,p} |x|^{-\frac{\mathbf{N}}{p}} \|f\|_{\mathbf{L}^p(\mathbb{R}^N)}, \quad 1 \leq p < \infty. \end{aligned} \quad (3.22)$$

Recall $\mathbf{U}_1 f$ defined in (3.5). Note that $0 \leq \frac{|y|}{|x|} \leq \frac{1}{2}$ implies $\frac{1}{2}|x| \leq |x| - |y| \leq |x - y|$. Let $f, g \geq 0$ and $f \in \mathbf{L}^p(\mathbb{R}^N)$, $g \in \mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^N)$. We have

$$\begin{aligned} \int_{\mathbb{R}^N} \mathbf{U}_1 f(x) g(x) dx &= \int_{\mathbb{R}^N} \left\{ \int_{|y| \leq \frac{1}{2}|x|} \frac{f(y)g(x)}{|x|^\gamma |x-y|^{\mathbf{N}-\alpha} |y|^\delta} dy \right\} dx \\ &\lesssim \int_{\mathbb{R}^N} \left\{ \int_{|y|<|x|} \frac{f(y)g(x)}{|x|^{\gamma+\mathbf{N}-\alpha} |y|^\delta} dy \right\} dx \\ &= \int_{\mathbb{R}^N} |x|^{\alpha-(\gamma+\delta)} g(x) \left\{ |x|^{-\mathbf{N}+\delta} \int_{|y|<|x|} f(y) |y|^{-\delta} dy \right\} dx \\ &= \int_{\mathbb{R}^N} |x|^{\alpha-(\gamma+\delta)} g(x) \mathbf{V}_\delta f(x) dx \\ &\leq \left\{ \int_{\mathbb{R}^N} |x|^{[\alpha-(\gamma+\delta)]q} (\mathbf{V}_\delta f)^q(x) dx \right\}^{\frac{1}{q}} \|g\|_{\mathbf{L}^{\frac{q}{q-1}}(\mathbb{R}^N)} \quad \text{by Hölder inequality.} \end{aligned} \quad (3.23)$$

Let $\frac{\alpha}{N} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{N}$, $1 \leq p < q < \infty$. We find

$$\begin{aligned}
& \left\{ \int_{\mathbb{R}^N} |x|^{[\alpha-(\gamma+\delta)]q} (\mathbf{V}_\delta f)^q(x) dx \right\}^{\frac{1}{q}} \\
& \leq \left\{ \int_{\mathbb{R}^N} |x|^{[\alpha-(\gamma+\delta)]q} |x|^{-N[\frac{q}{p}-1]} \|f\|_{L^p(\mathbb{R}^N)}^{q-p} (\mathbf{V}_\delta f)^p(x) dx \right\}^{\frac{1}{q}} \quad \text{by (3. 22)} \\
& = \|f\|_{L^p(\mathbb{R}^N)}^{1-\frac{p}{q}} \left\{ \int_{\mathbb{R}^N} (\mathbf{V}_\delta f)^p(x) dx \right\}^{\frac{1}{q}} \\
& \leq \mathfrak{C}_{\delta p} \|f\|_{L^p(\mathbb{R}^N)} \quad \text{by (3. 19).}
\end{aligned} \tag{3. 24}$$

From (3. 23)-(3. 24), we conclude

$$\|\mathbf{U}_1 f\|_{L^q(\mathbb{R}^N)} \leq \|f\|_{L^p(\mathbb{R}^N)}, \quad 1 \leq p < q < \infty. \tag{3. 25}$$

Consider

$$\mathbf{V}_\gamma g(x) = |x|^{-N+\gamma} \int_{|y|<|x|} |y|^{-\gamma} g(y) dy, \quad \gamma < \frac{N}{q}. \tag{3. 26}$$

We claim

$$\|\mathbf{V}_\gamma g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)} \leq \mathfrak{C}_{\gamma q} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}, \quad 1 < q < \infty. \tag{3. 27}$$

Write

$$\mathbf{V}_\gamma g(x) = \int_{\mathbb{R}^n} \Omega(|x|, |y|) g(y) dy, \quad \Omega(u, v) = \begin{cases} u^{-N+\gamma} v^{-\gamma} & \text{if } v < u \\ 0 & \text{otherwise.} \end{cases} \tag{3. 28}$$

Observe that Ω in (3. 28) is homogeneous of degree $-N$. Moreover,

$$\int_0^\infty \Omega(1, t) t^{\frac{N}{q}-1} dt = \int_0^1 t^{\frac{N}{q}-\gamma-1} dt < \infty \tag{3. 29}$$

provided by $\gamma < \frac{N}{q}$. Lemma 2.1 implies (3. 27).

On the other hand, $\mathbf{V}_\gamma g$ defined in (3. 26) satisfies

$$\begin{aligned}
\mathbf{V}_\gamma g(y) &= |y|^{-N+\gamma} \int_{|x|<|y|} |x|^{-\gamma} g(x) dx \\
&\leq |y|^{-N+\gamma} \left\{ \int_{|x|<|y|} |x|^{-\gamma q} dx \right\}^{\frac{1}{q}} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)} \quad \text{by Hölder inequality} \\
&\leq \mathfrak{C}_{\gamma q} |y|^{-N(\frac{q-1}{q})} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}.
\end{aligned} \tag{3. 30}$$

Recall $\mathbf{U}_2 f$ defined in (3. 6). Note that $\frac{|y|}{|x|} \geq 2$ implies $\frac{1}{2}|y| \leq |y| - |x| \leq |x - y|$. We have

$$\begin{aligned} \int_{\mathbb{R}^N} \mathbf{U}_2 f(x) g(x) dx &= \int_{\mathbb{R}^N} \left\{ \int_{|y| \geq 2|x|} \frac{f(y)}{|x|^\gamma |x - y|^{N-\alpha} |y|^\delta} dy \right\} g(x) dx \\ &\lesssim \int_{\mathbb{R}^N} \left\{ \int_{|y| > |x|} \frac{f(y) g(x)}{|x|^\gamma |y|^{N-\alpha+\delta}} dy \right\} dx. \end{aligned} \quad (3. 31)$$

By using (3. 31) and Tonelli's theorem, we have

$$\begin{aligned} \int_{\mathbb{R}^N} \mathbf{U}_2 f(x) g(x) dx &\lesssim \int_{\mathbb{R}^N} |y|^{\alpha-(\gamma+\delta)} f(y) \left\{ |y|^{-N+\gamma} \int_{|x| < |y|} g(x) |x|^{-\gamma} dx \right\} dy \\ &= \int_{\mathbb{R}^N} |y|^{\alpha-(\gamma+\delta)} f(y) \mathbf{V}_\gamma g(y) dy \\ &\leq \|f\|_{L^p(\mathbb{R}^N)} \left\{ \int_{\mathbb{R}^N} |y|^{[\alpha-(\gamma+\delta)] \frac{p}{p-1}} (\mathbf{V}_\gamma g)^{\frac{p}{p-1}}(y) dy \right\}^{\frac{p-1}{p}} \quad \text{by Hölder inequality.} \end{aligned} \quad (3. 32)$$

Let $\frac{\alpha}{N} = \frac{1}{p} - \frac{1}{q} + \frac{\gamma+\delta}{N}$, $1 \leq p < q < \infty$. For $p = 1$, we find

$$\begin{aligned} \| |y|^{\alpha-(\gamma+\delta)} \mathbf{V}_\gamma g(y) \|_{L^\infty(\mathbb{R}^N)} &\leq \mathfrak{C}_{\gamma \ q} |y|^{\alpha-(\gamma+\delta)} |y|^{-N(\frac{q-1}{q})} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)} \quad \text{by (3. 30)} \\ &= \mathfrak{C}_{\gamma \ q} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}. \end{aligned} \quad (3. 33)$$

For $p > 1$, write

$$\begin{aligned} &\left\{ \int_{\mathbb{R}^N} |y|^{[\alpha-(\gamma+\delta)] \frac{p}{p-1}} (\mathbf{V}_\gamma g)^{\frac{p}{p-1}}(y) dy \right\}^{\frac{p-1}{p}} \\ &= \left\{ \int_{\mathbb{R}^N} |y|^{[\alpha-(\gamma+\delta)] \frac{p}{p-1}} (\mathbf{V}_\gamma g)^{[\frac{p}{p-1} - \frac{q}{q-1}]}(y) (\mathbf{V}_\gamma g)^{\frac{q}{q-1}}(y) dy \right\}^{\frac{p-1}{p}}. \end{aligned}$$

By using (3. 30) again, we have

$$\begin{aligned} |y|^{[\alpha-(\gamma+\delta)] \frac{p}{p-1}} (\mathbf{V}_\gamma g)^{[\frac{p}{p-1} - \frac{q}{q-1}]}(y) &\leq \mathfrak{C}_{\gamma \ q} |y|^{[\alpha-(\gamma+\delta)] \frac{p}{p-1}} |y|^{-N(\frac{q-1}{q}) [\frac{p}{p-1} - \frac{q}{q-1}]} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}^{\frac{p}{p-1} - \frac{q}{q-1}} \\ &= \mathfrak{C}_{\gamma \ q} |y|^{N[\frac{1}{p} - \frac{1}{q}] \frac{p}{p-1} - N(\frac{q-1}{q}) [\frac{p}{p-1} - \frac{q}{q-1}]} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}^{\frac{p}{p-1} - \frac{q}{q-1}} \\ &= \mathfrak{C}_{\gamma \ q} \|g\|_{L^{\frac{q}{q-1}}(\mathbb{R}^N)}^{\frac{p}{p-1} - \frac{q}{q-1}}. \end{aligned} \quad (3. 34)$$

From (3. 32)-(3. 34), together with the $L^{\frac{q}{q-1}}$ -estimate in (3. 27), we conclude

$$\|\mathbf{U}_2 f\|_{L^q(\mathbb{R}^N)} \leq \|f\|_{L^p(\mathbb{R}^N)}, \quad 1 \leq p < q < \infty. \quad (3. 35)$$

Recall $\mathbf{U}_3 f$ defined in (3. 7). We have

$$\begin{aligned}\mathbf{U}_3 f(x) &= \int_{\frac{1}{2}|x| < |y| < 2|x|} f(y) \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} \left(\frac{1}{|y|}\right)^\delta dy \\ &\lesssim \int_{\mathbb{R}^N} f(y) \left(\frac{1}{|x-y|}\right)^{N-(\alpha-\gamma-\delta)} dy. \quad (\gamma + \delta \geq 0)\end{aligned}\tag{3. 36}$$

Note that $\frac{\alpha-\gamma-\delta}{N} = \frac{1}{p} - \frac{1}{q}$, $1 \leq p < q < \infty$. Define

$$\mathbf{I}_{\alpha-\gamma-\delta} f(x) = \int_{\mathbb{R}^N} f(y) \left(\frac{1}{|x-y|}\right)^{N-(\alpha-\gamma-\delta)} dy.\tag{3. 37}$$

For $p > 1$, we have

$$\|\mathbf{I}_{\alpha-\gamma-\delta} f\|_{L^q(\mathbb{R}^N)} \leq \mathfrak{C}_{p,q} \|f\|_{L^p(\mathbb{R}^N)}.\tag{3. 38}$$

This is a classical result due to Hardy, Littlewood and Sobolev [1]-[2].

Remark 3.1. For $p = 1$, we have $\mathbf{I}_{\alpha-\gamma-\delta}: L^1(\mathbb{R}^N) \longrightarrow L^{q,\infty}(\mathbb{R}^N)$ for $\frac{\alpha-\gamma-\delta}{N} = 1 - \frac{1}{q}$, $1 < q < \infty$. See chapter V of Stein [5].

Given $E \subset \mathbb{R}^N$, denote $\mathbf{vol}\{E\} = \int_E dx$. From (3. 36)-(3. 37) and **Remark 3.1**, we have

$$\begin{aligned}\lambda^q \mathbf{vol}\{x \in \mathbb{R}^N: \mathbf{U}_3 f(x) > \lambda\} &\leq \lambda^q \mathbf{vol}\{x \in \mathbb{R}^N: \mathbf{I}_{\alpha-\gamma-\delta} f(x) > \lambda\} \\ &\lesssim \|f\|_{L^1(\mathbb{R}^N)}^q, \quad \lambda > 0.\end{aligned}\tag{3. 39}$$

By replacing $f(x)$ with $f(x)|x|^\delta$ inside (3. 39), we obtain

$$\lambda \mathbf{vol}\left\{x \in \mathbb{R}^N: \int_{\frac{1}{2}|x| < |y| < 2|x|} f(y) \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} dy > \lambda\right\}^{\frac{1}{q}} \lesssim \int_{\mathbb{R}^N} f(x)|x|^\delta dx\tag{3. 40}$$

for every $\lambda > 0$.

Recall $\delta < 0$. Let $\delta_1 < \delta < \delta_2 < 0$ of which $\delta_i, i = 1, 2$ are close to δ . We find

$$\frac{\alpha}{N} = 1 - \frac{1}{q_i} + \frac{\gamma + \delta_i}{N}, \quad i = 1, 2\tag{3. 41}$$

for some $q_1 > q > q_2 > 1$.

By carrying out the same argument as (3. 36)-(3. 40), we simultaneously have

$$\lambda \mathbf{vol}\left\{x \in \mathbb{R}^N: \int_{\frac{1}{2}|x| < |y| < 2|x|} f(y) \left(\frac{1}{|x|}\right)^\gamma \left(\frac{1}{|x-y|}\right)^{N-\alpha} dy > \lambda\right\}^{\frac{1}{q_i}} \lesssim \int_{\mathbb{R}^N} f(x)|x|^{\delta_i} dx, \tag{3. 42}$$

$i = 1, 2$

for every $\lambda > 0$.

Next, we need to apply a Marcinkiewicz interpolation theorem of changing measures, due to Stein and Weiss [4].

Let $\mu_i, i = 1, 2$ be two absolutely continuous measures satisfying

$$\mu_i(E) = \int_E |x|^{\delta_i} dx, \quad i = 1, 2. \quad (3.43)$$

Define

$$\mu_t(E) = \int_E |x|^{\delta_1(1-t)} |x|^{\delta_2 t} dx, \quad \frac{1}{q_t} = \frac{1-t}{q_1} + \frac{t}{q_2}, \quad 0 \leq t \leq 1. \quad (3.44)$$

Theorem B: Stein and Weiss, 1958

Let $\mathbf{T}$ be a sub-linear operator, having the following properties:

(1) The domain of $\mathbf{T}$ includes $\mathbf{L}^1(\mathbb{R}^N, d\mu_1) \cap \mathbf{L}^1(\mathbb{R}^N, d\mu_2)$.

(2) If $f \in \mathbf{L}^1(\mathbb{R}^N, d\mu_i), i = 1, 2$, we have

$$\lambda \mathbf{vol} \{x \in \mathbb{R}^N : |\mathbf{T}f(x)| > \lambda\}^{\frac{1}{q_i}} \leq \mathfrak{C} \int_{\mathbb{R}^N} |f(x)| d\mu_i(x), \quad i = 1, 2. \quad (3.45)$$

Then,

$$\|\mathbf{T}f\|_{\mathbf{L}^{q_t}(\mathbb{R}^N)} \leq \mathfrak{C} \int_{\mathbb{R}^N} |f(x)| d\mu_t(x), \quad 0 < t < 1. \quad (3.46)$$

Remark 3.2. Theorem B in the more general setting of measurable spaces can be found in the paper by Stein and Weiss [4].

Recall $\frac{\alpha}{N} = 1 - \frac{1}{q} + \frac{\gamma+\delta}{N}$ and (3.41). There is a $0 < t < 1$ such that $\delta = (1-t)\delta_1 + t\delta_2$ and $\frac{1}{q} = \frac{1-t}{q_1} + \frac{t}{q_2}$. By using (3.42) and applying **Theorem B**, we obtain

$$\|\mathbf{U}_3 f\|_{\mathbf{L}^q(\mathbb{R}^N)} \leq \|f\|_{\mathbf{L}^1(\mathbb{R}^N)}. \quad (3.47)$$

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